

NetPractice IPv4 + Subnet Quick Reference

Fast lookup for CIDR, masks, block sizes, usable addresses, and common traps.

CIDR / mask lookup

CIDR	Subnet mask	Block	Usable hosts
/16	255.255.0.0	65,536	65,534
/17	255.255.128.0	32,768	32,766
/18	255.255.192.0	16,384	16,382
/19	255.255.224.0	8,192	8,190
/20	255.255.240.0	4,096	4,094
/21	255.255.248.0	2,048	2,046
/22	255.255.252.0	1,024	1,022
/23	255.255.254.0	512	510
/24	255.255.255.0	256	254
/25	255.255.255.128	128	126
/26	255.255.255.192	64	62
/27	255.255.255.224	32	30
/28	255.255.255.240	16	14
/29	255.255.255.248	8	6
/30	255.255.255.252	4	2

Valid mask-octet values

Mask octet	Bits
0	0
128	1
192	2
224	3
240	4
248	5
252	6
254	7
255	8

A normal subnet-mask octet must be one of these values. Example: **1** is invalid.

What the parts mean

IPv4

4 octets, each 0-255.

Example

96 . 224 . 118 . 132

Network bits

Identify the subnet.

Host bits

Identify the device inside that subnet.

CIDR

/24 = 24 network bits out of 32.

Rule

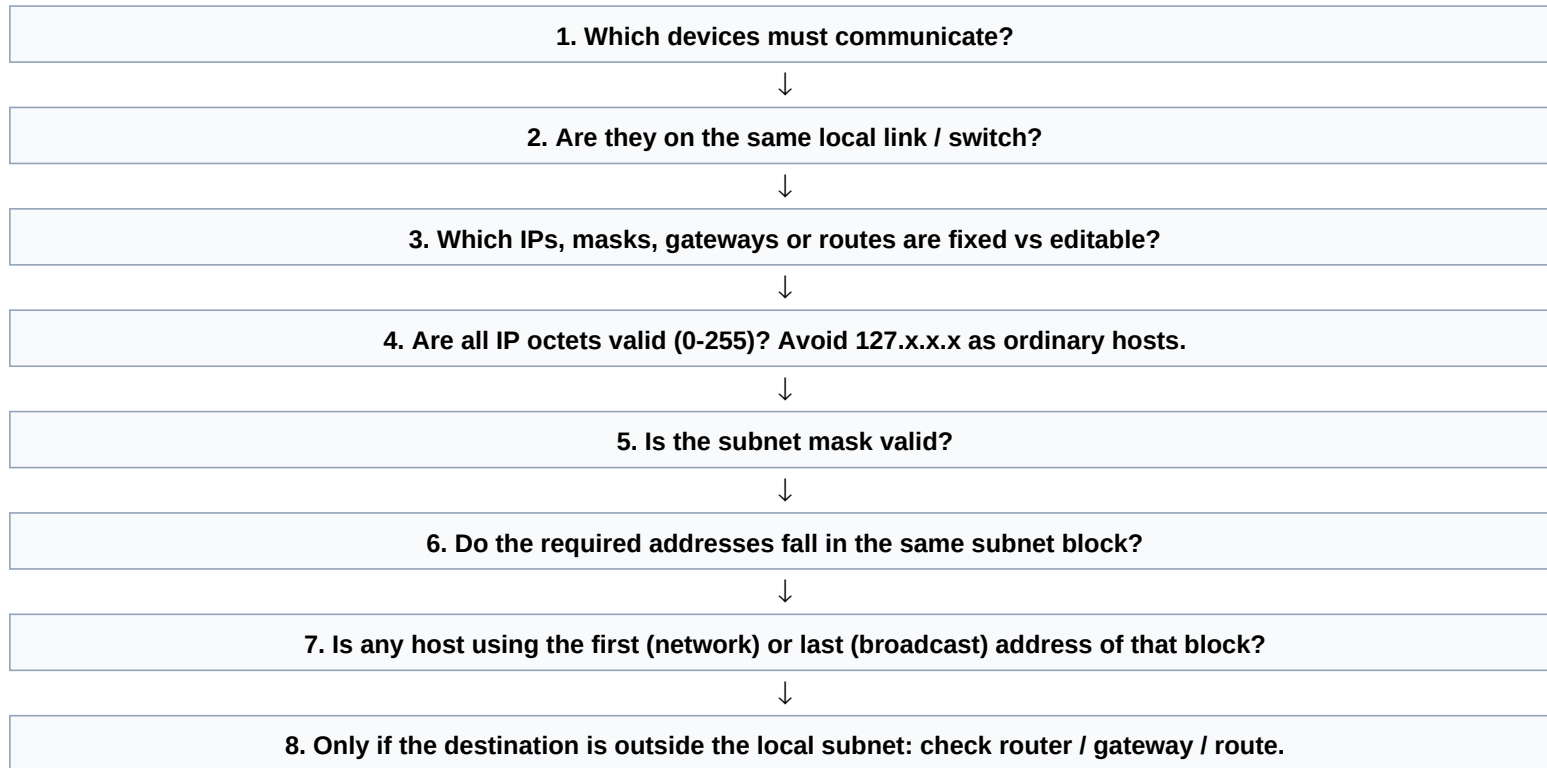
Higher /number = smaller subnet.

Block boundaries when the 4th octet is being subnetted

/25	0-127 128-255
/26	0-63 64-127 128-191 192-255
/27	0-31 32-63 64-95 96-127 128-159 160-191 192-223 224-255
/28	0-15 16-31 32-47 ... 240-255
/29	0-7 8-15 16-23 ... 248-255
/30	0-3 4-7 8-11 ... 252-255

Within a normal subnet block: first address = network address; last address = broadcast address; addresses in between = ordinary usable hosts.

NetPractice quick-check flow



Common traps

Value / range	Meaning
127.0.0.0/8	Loopback - do not use as ordinary hosts in NetPractice.
First address of subnet block	Network address - not an ordinary host.
Last address of subnet block	Broadcast address - not an ordinary host.
255.255.255.0	/24
255.255.0.0	/16

Tiny glossary

IP address - device address.
Subnet mask - decides network vs host portion.
Switch - connects devices on the same local network.
Router - connects different networks.
Gateway - router interface used to leave the local network.

Mental model: IP = house address. Subnet mask = neighborhood boundary. Same subnet = local communication. Different subnet = routing needed.